

MID TERM EXAMINATION

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 41 | Date: 13-03-2026

Total Questions: 10 | Answer all questions.

Name: _____

Roll No: _____

Student Sign: _____

Teacher Sign: _____

SECTION A: MCQs

1. What is one of the seven core design problems identified for Lexi's design? [2 Marks]
2. In Lexi's document structure, what is the common abstract class that serves as the base for all elements, from primitive characters to complex structures? [2 Marks]
3. AnswerUnderstand Describe the problem that the Strategy Pattern solves regarding formatting in Lexi, and how it achieves this solution. [5 Marks]
4. AnswerUnderstand In the context of supporting multiple window systems, explain the independent variation achieved by the Bridge Pattern between Lexi's windowing abstraction and its implementation. [5 Marks]
5. AnswerUnderstand Discuss how the combination of the Iterator and Visitor patterns enables analytical operations like spelling checking and hyphenation without changing the Glyph classes or using type-checking. [5 Marks]

SECTION B: SHORT ANSWERS

6. Which design pattern is used to add visual embellishments like borders and scroll bars dynamically without creating a combinatorial explosion of subclasses? [2 Marks]
7. AnswerUnderstand Explain how the Composite Pattern, using the Glyph abstract class, allows clients to treat individual objects and compositions of objects uniformly in Lexi's document structure. [5 Marks]
8. AnswerUnderstand How does the Abstract Factory Pattern help Lexi in supporting multiple look-and-feel standards without hard-coding class names throughout the application? [5 Marks]
9. AnswerUnderstand Describe how the Command Pattern provides support for multi-level undo/redon functionality for user operations in Lexi. [5 Marks]

- 10.** Answer Understand Analyze the core problem the Decorator Pattern solves for embellishing the UI and how it achieves a flexible solution compared to a simpler approach. **[5 Marks]**
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